

Box 1: BAM-trait based biodiversity model

We outline a coevolutionary IBM model to evaluate the effect of trait architecture (i.e., the covariation among biotic, abiotic and migration traits) on rescue and biodiversity dynamics (Figure 1a). We explore two trait architectures rooted in empirical and theoretical studies (Armbruster et al. 2014): the modular and the trait integration architectures. The modular architecture takes into account independent traits, i.e. each trait contributes independently to fitness (Figure 1b: Modular). The integration trait architecture incorporates the biotic trait as the main contributor to fitness with the abiotic and the migration traits depending on the evolution of the biotic trait (Fig 1b: Magic).

Our model considers a landscape with a network of patches connected by migration events and inhabited by resource and consumer populations comprising several individuals each. Biotic, abiotic and migration trait values are initially sampled from normal distributions for each individual phenotype i represented as $\mathbf{z} = [z_b, z_a, z_d]$. The local resource demography of species i is driven by the temporal dependent fitness function following

$$W_R(\mathbf{z}_{B_i})_x^t = 1/[1 + \exp(-\gamma(\mathbf{z}_{B_i x}^t - \hat{\mathbf{z}}_{B_{\text{forall}j} x}^t)^2)] , \quad (1)$$

$$W_R(\mathbf{z}_{A_i})_x^t = \exp[-\gamma(\mathbf{z}_{A_i x}^t - \hat{\mathbf{z}}_{A_i x}^t)^2], \quad (2)$$

$$W_R(\mathbf{z}_{M_i})_x^t = \exp[-\gamma(\mathbf{z}_{M_i x}^t - \hat{\theta}_{D_x})^2], \quad (3)$$

and for the consumers, the fitness functions are the following

$$W_C(\mathbf{z}_{B_i})_x^t = \exp[-\gamma(\mathbf{z}_{B_i x}^t - \mathbf{z}_{B_j x}^t)^2], \quad (4)$$

$$W_C(\mathbf{z}_{A_i})_x^t = \exp[-\gamma(\mathbf{z}_{A_i x}^t - \theta_{A_x})^2], \quad (5)$$

$$W_C(\mathbf{z}_{M_i})_x^t = \exp[-\gamma(\mathbf{z}_{M_i x}^t - \theta_{D_x})^2], \quad (6)$$

where $\mathbf{z}_{T_x}^t$ is the trait value T of phenotype i at time t and site x , θ_x is the multivariate fitness optimum, ω is the covariance matrix (Fig 1) (Lande, 1980; Melo and Marroig, 2014), and γ determines the interaction sensitivity to deviations from the biotic (i.e., coevolutionary selection strength), abiotic and dispersal optimum.

In this equation, demographic and trait evolutionary dynamics depend on the scenario of trait architecture being considered.

If the covariance matrix, ω , is diagonal, representing our modular scenario (Fig. 1), then there is no correlated stabilizing selection. If the covariance matrix is not diagonal and considering our magic scenario (Fig. 1), then there is correlated stabilizing selection with the biotic trait determining the selection on the other traits. The biotic optimum changes in time and we define it as the minimum and maximum mean trait distance across all populations in site x for the consumer and the resource biotic trait value, respectively. The dispersal optimum is given by the mean distance in the landscape (i.e., each site has coordinates from which we calculate the Euclidian distance between each pair of sites). Phenotypes with dispersal trait value near to this optimum maximize fitness for this trait only for the modular but not for the magic trait scenario. The abiotic optimum is taken initially as the mean of the distribution and it remains the same during the dynamics.

At each time step a certain percentage of individuals are selected randomly to die and are replaced by birth or migration. The time dependent fitness function in site x , $W(\mathbf{z})_x^t$, determines the mother of the offspring. To obtain the phenotypic value for the offspring, $z_o = [z_{ob}, z_{oa}, z_{od}]$, we add to each trait value of the mother the environmental deviation, $\mathbf{e}$, taken from a multivariate uniform distribution with range ± 0.001 in all dimensions (i.e., independent sampled values for the modular but correlated for the integration trait scenario). We run our model for 10000 replicates with 50 generations each for the modular and integration trait scenarios. To evaluate the importance of coevolution and migration as rescue mechanisms in modulating biodiversity dynamics, we explore how the formation of hot and cold spots of species persistence in local and regional biodiversity changes in a gradient of migration and coevolutionary selection strength (Figure x2).

Codes and outputs can be downloaded from <https://github.com/melian009/Cometa>.

1. Bibliography

- R. Lande. The genetic covariance between characters maintained by pleiotropic mutations. *Genetics*, 94: 203–215, 1980.
- D. Melo and G. Marroig. Directional selection can drive the evolution of modularity in complex traits. *Proceedings of the National Academy of the Sciences, USA.*, 112:470–475, 2014.